Paper 21 - MorphForge / PolyForge / MorphoniX

CQE-paper-21
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Read glyphs, numbers, shapes, and tokens as grid-swept ribbons whose bifurcations define folds and continuations.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-21.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-21\paper_kernel_manifest.json
- body: production\papers\CQE-paper-21\PAPER-BODY.md
- source: production\papers\CQE-paper-21\SOURCE.md
- formal: production\papers\CQE-paper-21\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-21\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-21\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-21.md

Paper 21 - MorphForge / PolyForge / MorphoniX

Abstract

Paper 21 defines the applied Forge reader. Its closed result is that an observed object can be converted into a grid-swept ribbon, encoded as a lossless symmetric-group word, accounted as morphon records, and landed in the 24-dimensional terminal template while every unfinished bridge remains explicit.

The formal receipt verifies a Rule-30 shell-2 ribbon through depth 4096. It contains 1569 shell-2 states and a 1568-step S3 word with zero round-trip mismatches. The morphonics ledger closes its schema with 5 morphons, 5 transforms, 5 projections, 3 accounting records, 3 bridges, 11 claims, 3 explicit failure records, and 5 passing morphon closure tests. The terminal landing form is the canonical `Niemeier:E8^3` composition tree: ambient dimension 24, root rank 24, and residue closed by required index.

The paper therefore proves the MorphForge reading contract. It does not by itself prove cross-medium equivalence, a Mandelbrot boundary theorem, a Leech lattice construction, TF1 closure, biological closure, material closure, or CAD closure. Those items remain open obligations unless later papers attach domain-specific verifiers.

Definitions

An observation event begins when an observer chooses a centroid or object to enumerate. Paper 21 treats that choice as a ribbon source: a sequence of local windows that may be read as left, center, and right boundary states.

A ribbon is a finite ordered list of those windows. A shell-2 ribbon is the subtrajectory selected by the local shell rule used by the included Rule-30 chart codec. A fold is a non-identity transition in S3. A morphon is a ledger row that records a source, transform, projection, accounting link, and claim status for a readable object. A terminal landing template is the 24-dimensional composition tree used to place the reading into the lattice suite without overstating the placement.

Claims

- The chart codec closes a lossless ribbon encoding.

The verifier reports `status = pass`, `round_trip_mismatches = 0`, and `first_mismatch = null` for the Rule-30 shell-2 trajectory through depth 4096. The shell-2 ribbon has length 1569, and its S3 word has length 1568, exactly one transition per adjacent pair of shell states.

- Folds are observable as non-identity S3 steps.

The receipt records 494 identity self-loops and 1074 non-identity transitions. The non-identity transitions observed in this run are transposition-class steps: 197 `(1 2)`, 594 `(1 3)`, and 283 `(2 3)` events. The two 3-cycle counts are zero in this receipt. Paper 21 therefore proves a fold classifier for this ribbon, not a universal theorem that every future ribbon has the same element distribution.

- The morphonics model closes as an accounting substrate with carried gaps.

The morphonics verifier returns `pass_with_open_gaps`. Schema status is `pass`; all five morphon closure tests pass; and the three open failure labels are explicit: `PENDING_IMPORT`, `MISSING_MORPHISM`, and `PENDING_MEASUREMENT`.

- The terminal landing form is a 24-dimensional template, not a Leech construction.

The terminal tree is `Niemeier:E8^3`, with ambient dimension 24, root rank 24, three component-action branches, 24 compact involution slots, and residue closed by required index. This proves that the MorphForge reader can land its accounting in the 24-dimensional lattice package. It does not prove a Leech import unless the missing Golay or Construction-A data are supplied.

- Applied-domain claims inherit the substrate and the residue.

GraphSTX calendar flows, FridgeForge recipe flows, CADForge/WireBlock design flows, metamaterial sketches, TF1 examples, biological examples, and other product-facing uses may all use the Paper 21 reader. They do not become closed scientific claims from that inheritance alone. Each must attach a domain receipt or carry an open-obligation label.

Theorem 21

The MorphForge reader is a valid CQE applied-reader kernel exactly when it returns three artifacts for a chosen observation event: a lossless ribbon word, a morphon accounting ledger with explicit closure and failure statuses, and a terminal landing template whose strength is not overstated.

Proof

Run `verify_morphforge_ribbon.py`. The first check verifies that the chart codec round-trips the shell-2 trajectory with no mismatches. Because the encoded word has one element for every adjacent pair in the shell-2 ribbon, the word is not a summary or illustrative sketch; it is a reversible reading of the selected subtrajectory.

The next checks classify the transition content of that word. The identity self-loops preserve local state. The 1074 non-identity steps are the fold events in this receipt. Since the verifier also reports the S3 element counts, a reviewer can falsify the fold claim by finding a mismatch between the word, the decoded ribbon, and the reported counts.

The morphonics checks examine the ledger. The model is accepted only as `pass_with_open_gaps`; unresolved bridges are not silently promoted into solved claims. Every morphon has a passing closure test, and the missing Leech import, expanded morphism witnesses, and TF1 measurement remain named failure records.

The final check lands the reading in the `Niemeier:E8^3` terminal tree. The receipt confirms the 24-dimensional rank and residue condition needed for the suite-level placement. Because the receipt identifies the evidence level as a template, the proof does not smuggle in a completed Leech construction or a domain-specific experimental proof.

Therefore Paper 21 proves the applied Forge reading and accounting kernel while preserving the exact boundary of what is not yet closed.

Receipt

The formal receipt is generated at `production/formal-papers/CQE-paper-21/morphforge_ribbon_receipt.json`.

The paper passes when every listed check passes and the remaining obligations are carried in the receipt rather than omitted from it.

Falsification Conditions

Paper 21 fails if the shell-2 word does not decode back to the selected ribbon, if the word length differs from the number of adjacent shell transitions, if the reported S3 counts disagree with the encoded word, if a morphon lacks accounting closure, if an open bridge is promoted without a verifier, or if the terminal landing is represented as stronger than its receipt.

Suite Role

Paper 20 supplies the ledger admission rule. Paper 21 supplies the applied reader used by later product, material, biological, and design papers. Papers 22 through 24 may inherit the ribbon, fold, morphon, and terminal-template discipline, but they must add their own domain verifiers before claiming closure.

Conclusion

MorphForge, PolyForge, and MorphoniX are the same formal move at different scales: choose the observation event, read the local windows as a reversible ribbon, account the transformation in the morphon ledger, and land only the verified result into the lattice suite. The power of the method is not that it declares every bridge solved. The power is that it makes each bridge visible, portable, and testable.